| | |
|---|---|
| Title: | Lab 6: Form Layout Techniques |
| Special Notes / Additional Resources: | Only one completed database is used for this lab. All parts of the lab should be completed in the same database which can then be uploaded as your submission.<br><br>A submission template for you to place screenshots in has also been provided with the lab and is required as part of your submission. |

# Part A: Preparing Table Properties for Form Development [ 3 marks ]

**Overview**

- This lab provides essentials on how to build nicely presented and aesthetic forms in MS Access.
- Download the starting point database for the lab before commencing these instructions and rename it with your name in the filename (e.g., Sousa-12345-Lab5.accdb).

**About the Sample Database**

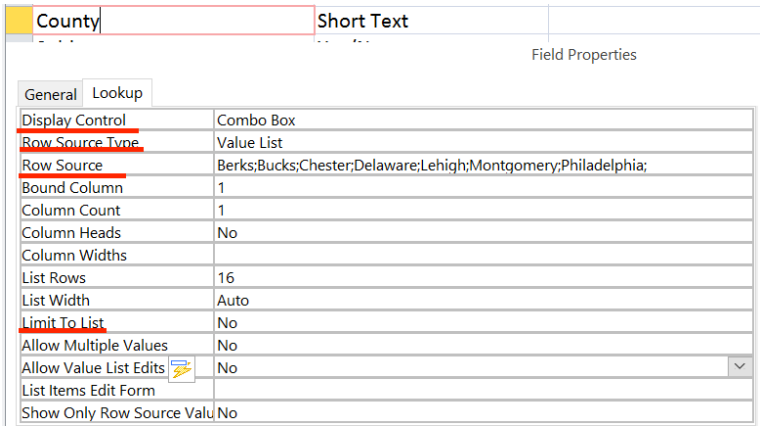
- Review the relationships in the database to familiarize yourself with this sample real estate data model which allows agents to keep track of listings within their own company.
- A unique feature of this application is that agents can also list "extras" that the owners are also willing to sell with the property. This includes things like furniture and appliances. Relatively very few property listings have extras listed.
- We have already configured combo boxes in this database for all of the important foreign keys (Agent and Subdivision) to provide better ease of use.

**Instructions**

1. **[3 marks]** Before we create forms for this application, we will review the table properties and makes some changes that will make some aspects of the form design easier.
   a. Open the Properties table in design view. Notice we have a field called "Sold" which is a yes/no field. Yes/no fields work best as checkboxes rather than textboxes.

      i. In the "Lookup" properties of the County field, we want to add the convenience of frequently used counties but still allow free-form typing of values. Since most properties are in one of 7 counties, we'll create a combo box for them.

      Set the Lookup Properties as shown below (red/underlined items should be set as shown, other properties can be left in their default values).



b. We'll make similar adjustments to the fields "Beds" and "Baths", however, we'll limit the choices to the list. This simplifies data validation since our users would be prevented from entering negative values or unreasonably large values. We'll allow 0 – 10 for Beds and Baths (zero could be valid if an empty parcel of land was being listed for example).

Properties for Beds

| | |
|---|---|
| Beds | Number |
| Baths | Number |
| Address | Short Text |
| SubDivisionID | Number |
| County | Short Text |

General | Lookup

| | |
|---|---|
| Display Control | Combo Box |
| Row Source Type | Value List |
| Row Source | 0;1;2;3;4;5;6;7;8;9;10 |
| Bound Column | 1 |
| Column Count | 1 |
| Column Heads | No |
| Column Widths | |
| List Rows | 16 |
| List Width | Auto |
| Limit To List | Yes |

Properties for Baths

| | |
|---|---|
| Baths | Number |
| Address | Short Text |
| SubDivisionID | Number |
| County | Short Text |
| Sold | Yes/No |

General | Lookup

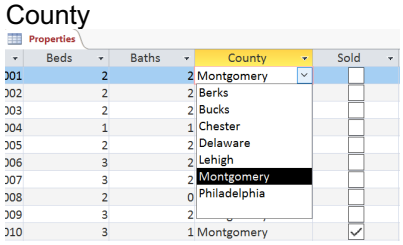
| | |
|---|---|
| Display Control | Combo Box |
| Row Source Type | Value List |
| Row Source | 0;1;2;3;4;5;6;7;8;9;10 |
| Bound Column | 1 |
| Column Count | 1 |
| Column Heads | No |
| Column Widths | |
| List Rows | 16 |
| List Width | Auto |
| Limit To List | Yes |

c. Close the design view of the table and save the changes. Before designing a form, we can test the effect of the property changes by opening the table in a datasheet view (just double click the table to open it). Try editing a few of the rows to ensure the drop down lists are working as expected. (partial sample shown below).

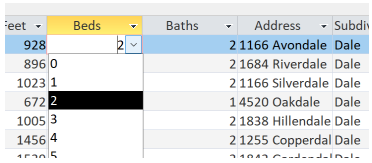
Screenshots 1,2, and 3

Take three screenshots showing the drop down list functioning for each of: county, beds, and baths drop down list and place them in the submission template

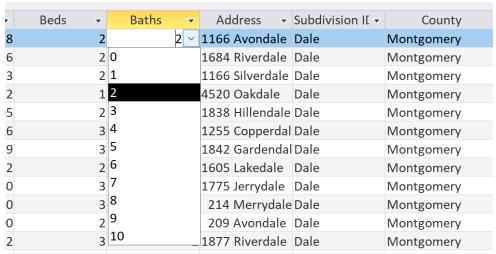
County




Beds




Baths



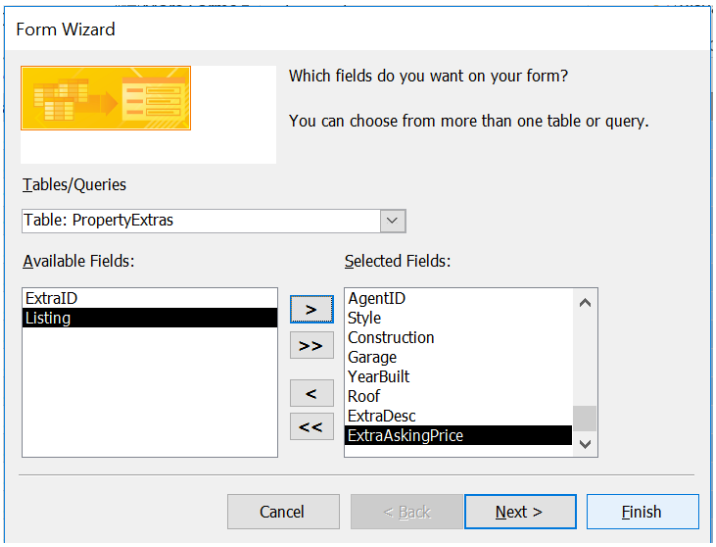
# Part B: Building a Two-Table Form and Improving Layout/Appearance [12 marks]

### Overview

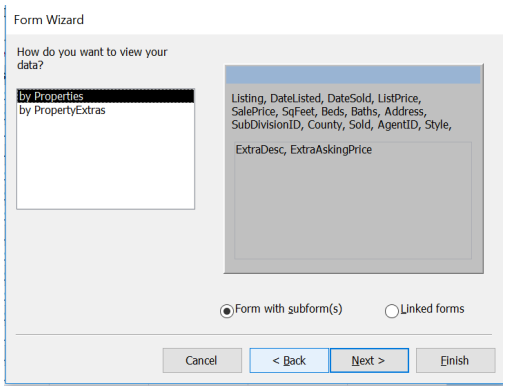
- The number of fields for the property listing makes it ideal for this exercise as we need to provide a more effective organization and layout of the information.

### Instructions

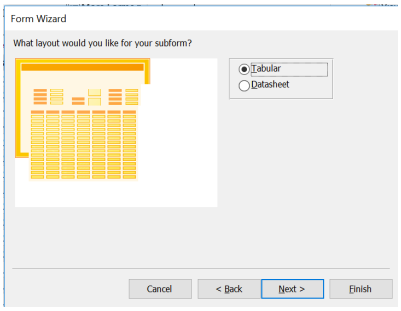
1. **[2 marks]** Create a new form using the Design Wizard.
   At step 1 in the wizard choose <u>all</u> the fields from the Property table and the ExtraDesc and ExtraAskingPrice fields from the PropertyExtras table.



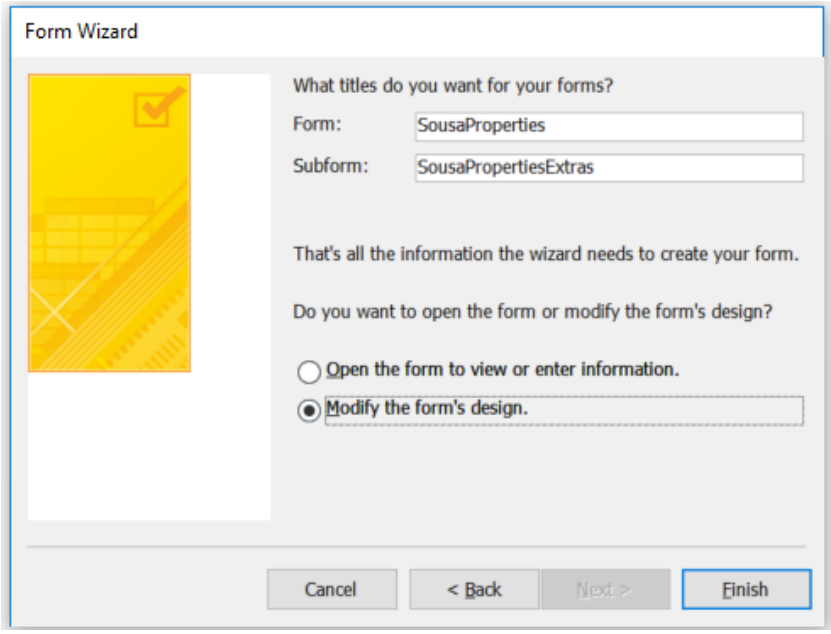

   a. At step 2, choose “by properties” and “Form with Subform” to layout the information in a two-tiered approach then click next.



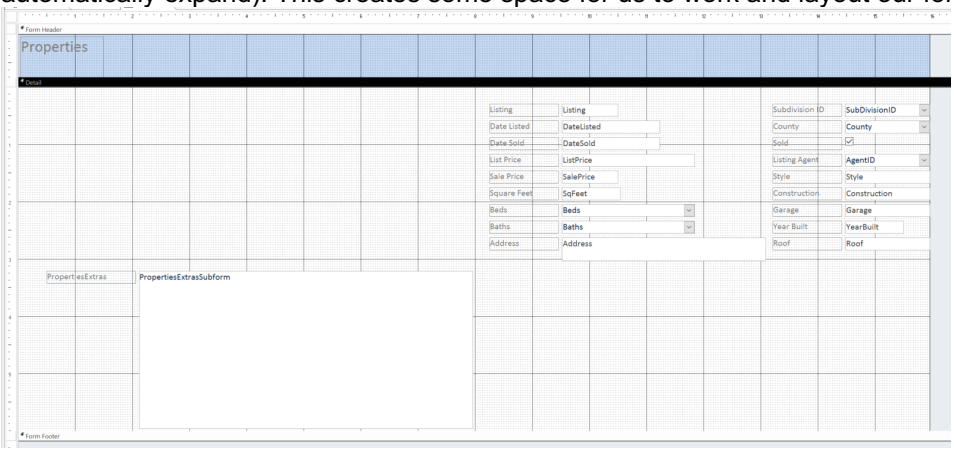
b. At step 3 in the wizard, we'll choose tabular layout for the subform. Tabular layouts give us more design and layout control in general. The datasheet view is what we see by default in MS Access when we open tables for editing.



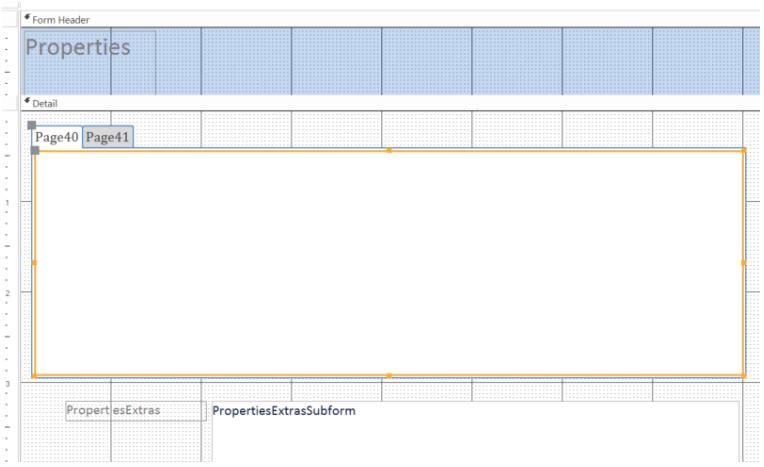

c. In the final step of the wizard, we can specify the names of the forms. A best practice is to name the form and subform with the same prefix so they appear as a pair/side-by-side. Name the subforms as shown and choose to open the form in design view before clicking finish. Prefix the form name with your last name (example shown)



2. **[ 4 marks ]** Using a Tab Control to provide a better arrangement of fields is an excellent technique that maximizes screen real-estate when you have a lot of attributes to include.
   a. We'll start by selecting all the property fields (drag an area around them to move all of them at once) and moving them temporarily over to the right (the canvas will automatically expand). This creates some space for us to work and layout our form.

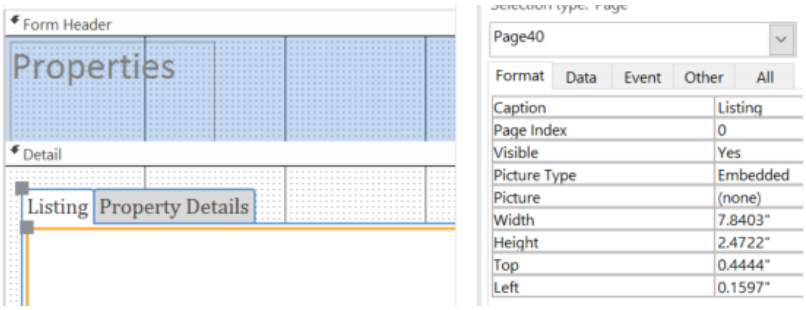



   b. From the ribbon, add a new tab control to the form. The tab control looks like tabs on a binder or a file folder. . When we create the tab control, we'll make it roughly as wide as the lower part of the form.

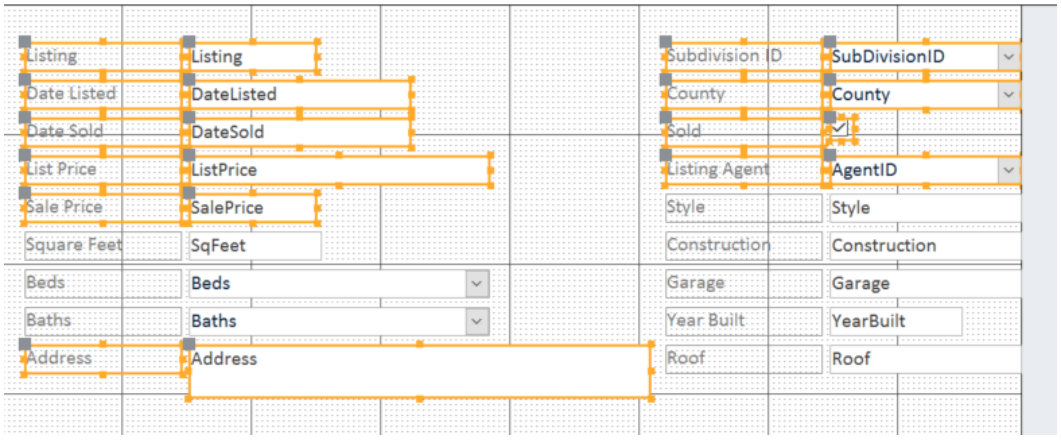


c. The tabs on a tab control are called pages. By default, we have two pages which is sufficient for this exercise (pages can be added or removed by right clicking the tab control).

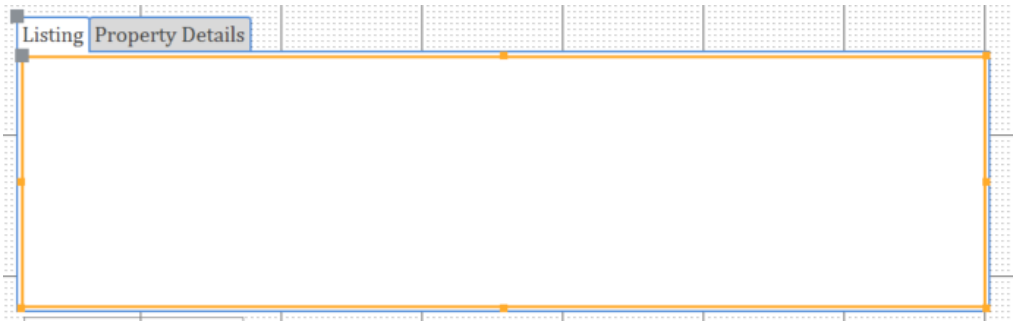
We need to give the pages better captions. Click on each one and in the properties window (on the right) you'll find the caption property under "Format". Set the first one to Listing, and the Second one to "Property Details". If the property window is not shown, you can open it by clicking the tab control and choosing "Properties" from the right click menu.



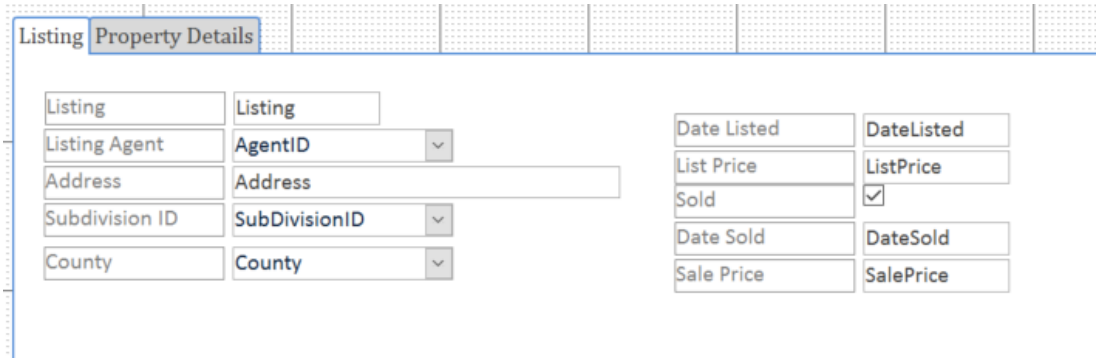

d. Now we'll cut and paste the fields we want to arrange onto each page. While holding control on the keyboard we can click the individual fields and labels we want. The first tab will hold information about the sale itself. Select the fields shown and cut them (Ctrl-X).



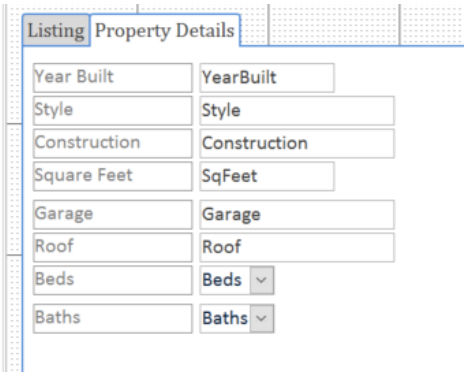
e. Select the first tab that we want to paste them on. We will need to click the tab twice so that there is an orange boundary around the page. If you don't do this, the controls, once pasted, will not be "inside" the page.




f. After pasting the fields, we'll arrange them roughly as shown. We'll also resize address so it appears more consistent (address is a long text field and by default is shown with a multi-line display). We can also resize some of the other controls so they are tidier.

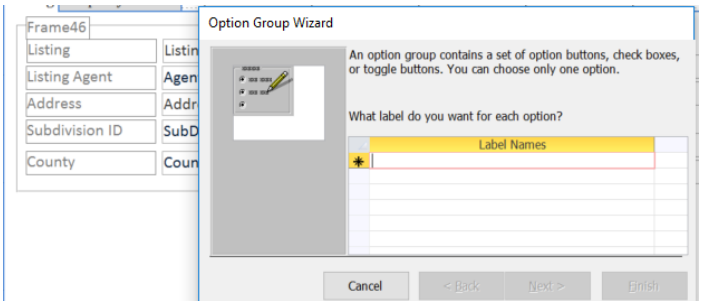



g. Next, we'll do the same to the remaining fields but paste them on the property details page. We'll also arrange and resize them as shown.

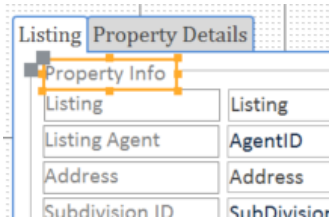


3. **[ 2 marks ]** Next we'll create an option group. Option groups are actually intended to hold option buttons (also known as radio buttons) but are also handy as a way to visually group related information on an interface.

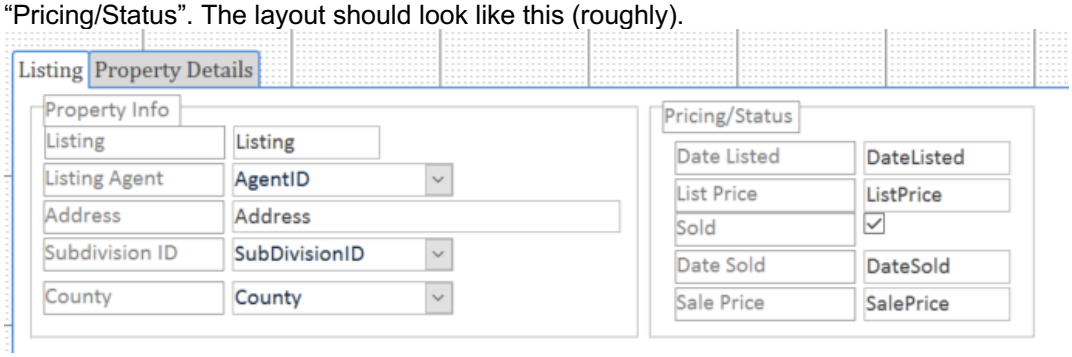
   a. Create an option group that surrounds the first set of fields on the Listing tab. If the control wizard pops up, we can simply cancel it since we don't want to use it for those purposes.



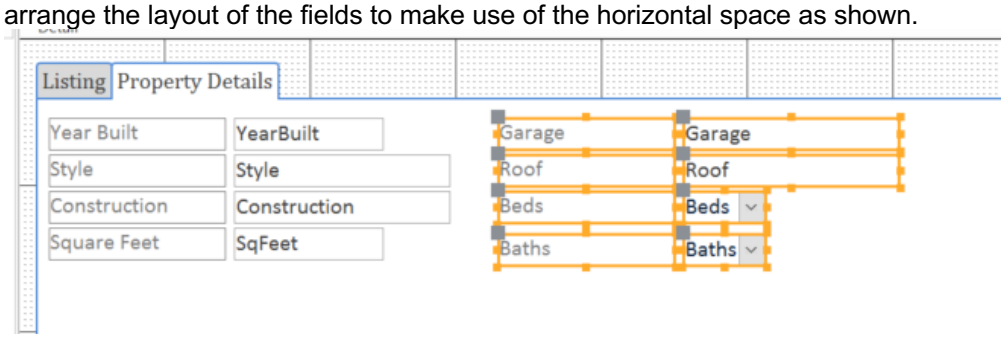

   b. Click on the label of the option group (default will be "Framexx") and set its caption property to "Property Info". The label will need to be wider to fit this text so resize it as shown by clicking/dragging on its right border.



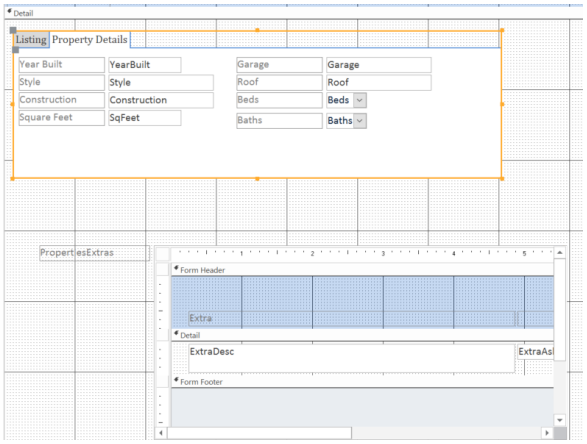

   c. Create a second option group / frame for the other group of fields and label it as "Pricing/Status". The layout should look like this (roughly).



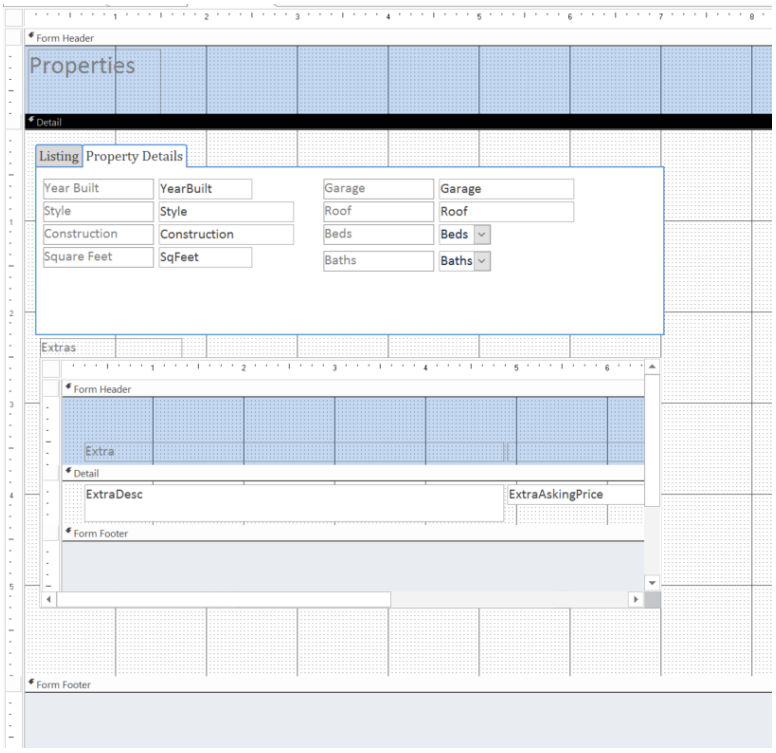
d. Now we'll examine the second tab. Notice that the current arrangement we have leaves a lot of un-used space which would require the tab to be taller than needed. We'll re-arrange the layout of the fields to make use of the horizontal space as shown.




e. Now we can re-size the overall tab control to make it as small as possible. Click/drag the controls lower right hand corner and shrink it down to size.



4. **[2 marks]** Final Refinements/Layout
   a. We'll finish our layout by improving the relative arrangement of the "Extras" subform. We'll change the caption of the subform to simply "Extras Available for Sale" and move the label above the subform. We'll widen the subform for aesthetics.



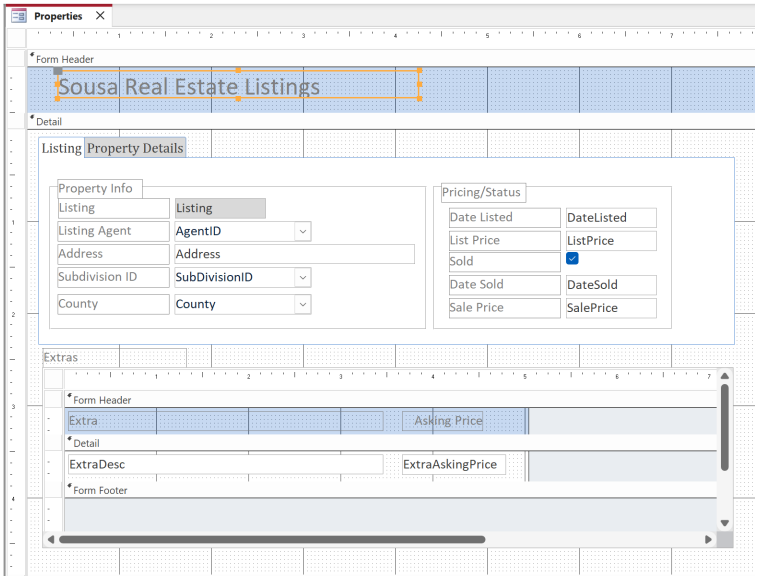
b. [**2 marks**] We'll now refine our layout as follows:

**Subform**

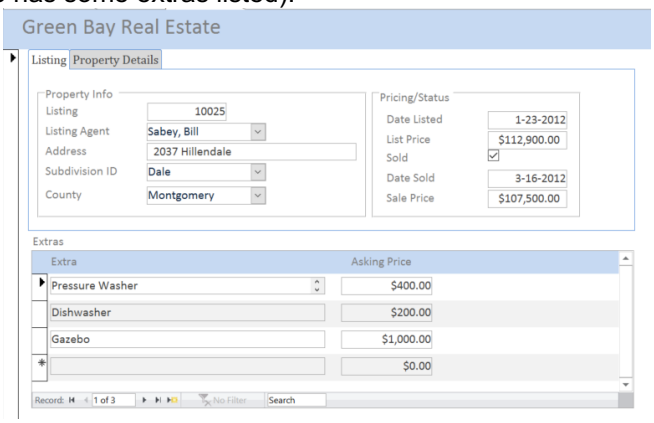
i. Move the field labels up to the top of the header.
ii. Reduce the height of the blue shaded header (click/drag on the dividing region labeled "Detail" upward).
iii. Resize the fields so that they fit better.
iv. Eliminating any extra vertical space in the "detail" section by ensuring the fields are tight against the "Detail" bar and then dragging the "Form Footer" bar upward to eliminate space.
v. Reducing the overall height of the subform (most properties will only have a few extras so it needn't be too tall)

**Form Title**

vi. One final step is to give the form a title that includes your name. The final design and refined layout appears as follows.

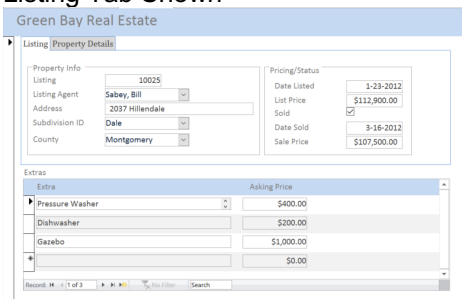
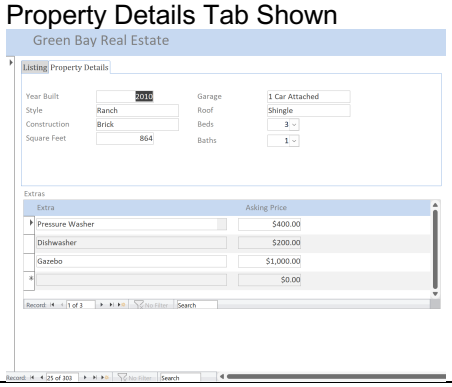



c. Test the form by opening it in form view. Navigate to record #25 using the record navigation buttons at the bottom of the form to see how the "extras" looks (2037 Hilendale has some extras listed).




d. If your form fields are not sized well for the data, you can change the view to "Layout View". Layout view is similar to the design view but allows you to resize form elements while viewing the data present in the database.

## Screenshot Preparation

| | |
|---|---|
| Screenshot 4 | Take a screenshot showing your final form showing record number 25 (2037 Hillendale). Your final design does not need to be a perfect match pixel-per-pixel but needs to show all the refinements from the instructions above.<br><br>Listing Tab Shown<br>  <br> |
| Screenshot 5 | Property Details Tab Shown<br>  <br> |

Hand In:

- The lab submission template containing your screenshots as evidence of completion.
- Your Access database which can be used to confirm completion of the lab or if needed during marking.